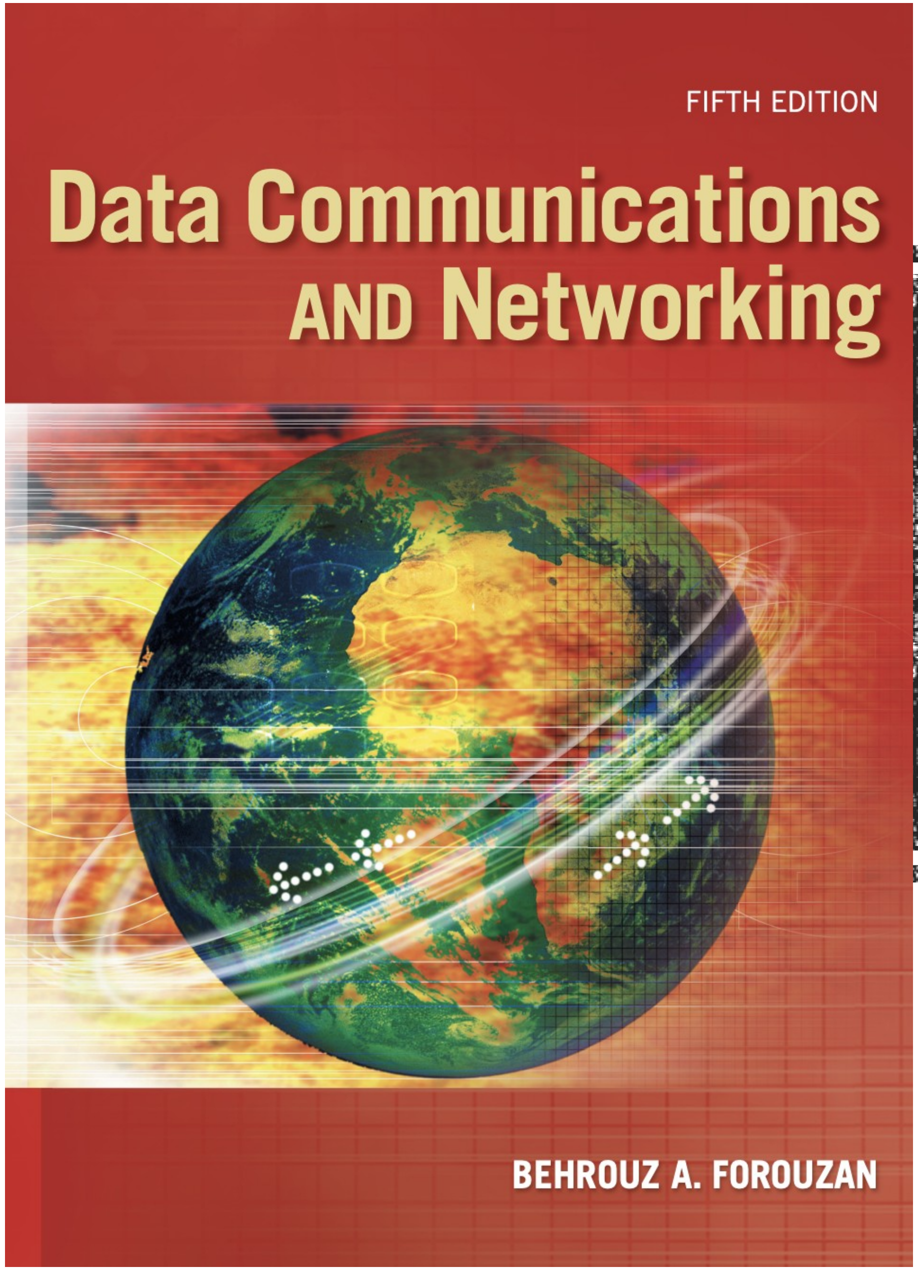


Chapter 9

Introduction to Data-Link Layer

The Role of the Data-Link Layer

No **define protocols** for the data-link or physical layers.

- These layers are managed by individual network technologies (wired/wireless).
- Provide foundational services to upper layers.

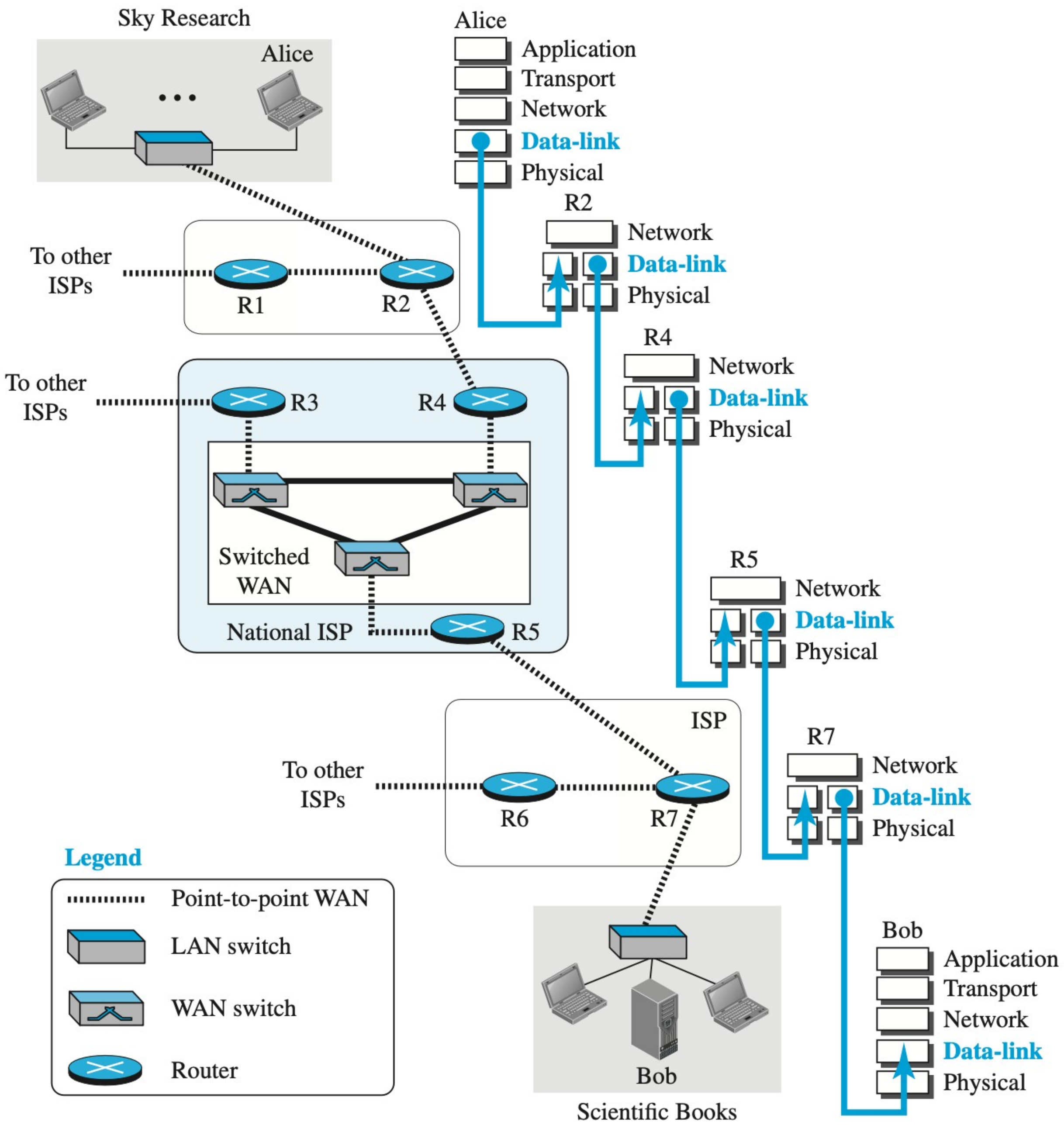
Section 1: Overview of the Data-Link Layer

- **Links** and **nodes**
- **Services provided** by the data-link layer
- Two **sublayers**: MAC and LLC

Section 2: Link-Layer Addressing

- The necessity for addressing at the data-link layer
- Three types of link-layer addresses
- Address Resolution Protocol (ARP)
  - Mapping of network layer addressing and link layer addressing
  - Locating next hop link layer address

Figure 9.1: Communication at the data-link layer



- The Internet consists of **multiple networks** connected by **routers or switches**
- For each network, a data link connection is required.
- If a device is connected to multiple networks, then that device is involved with multiple data link layer.
  - Only one data-link layer is involved in Alice and Bob
  - Two data-link layers are involved at each router. one for incoming, one for outgoing.

Nodes and Links

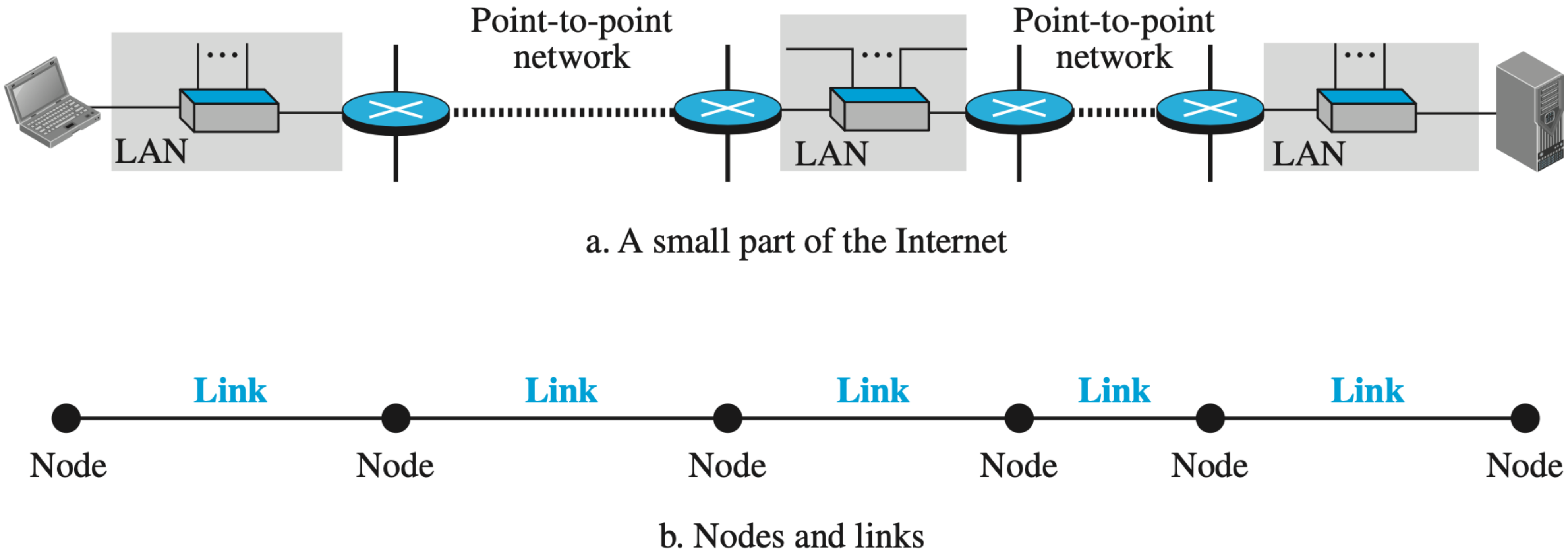


Figure 9.2: Nodes and Links

- Communication at the data-link layer is node-to-node.
- Two end hosts (Source and Destination) and the routers are ***nodes*** and the networks in between as ***links***.

Roles of Data Link Layer

- In between Physical and Network layers.
- Provides services to network layer, uses services of physical layer.

Scope of Data Link Layer

- Node-to-Node Delivery
- Responsible for sending a **datagram to the next node** in the path
- Uses **encapsulation** (sender) and **decapsulation** (receiver)

Encapsulation and Decapsulation

Source Host:

- Data-link layer encapsulates datagram into a **frame**
- Frame sent to the **router**

Router:

- **Decapsulates** the received frame
- **Encapsulates** the datagram into a new frame
- Sends the new frame to **destination host**
- *Note:* Router typically has **3 data-link layers** (one per physical link)

Destination Host:

- Data-link layer **decapsulates** the frame to extract the datagram

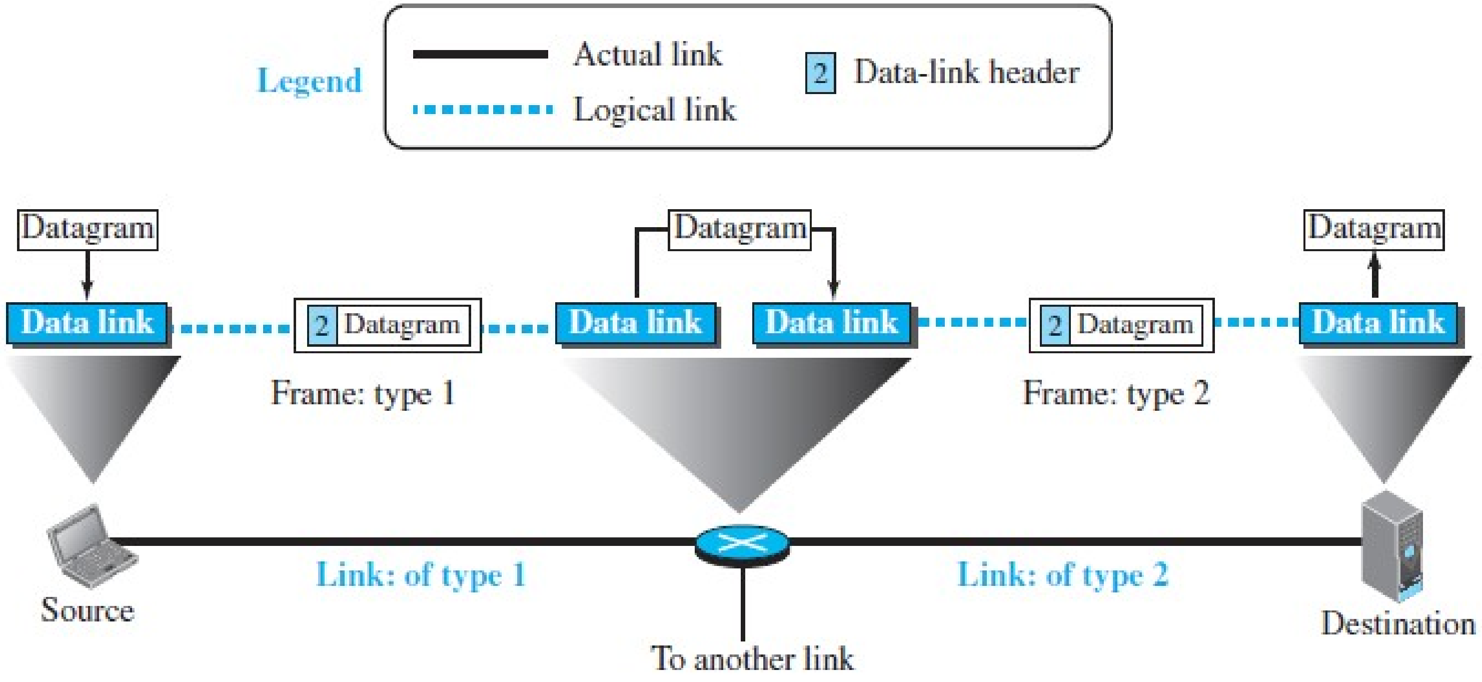


Figure 9.3: A communication with only three nodes

Services of Data Link Layer

- **Framing**
  - Encapsulates network-layer datagram into a frame
  - Adds header and trailer
  - Format varies by protocol
  - Ensures proper data boundaries

A packet at the data-link layer is normally called a ***frame***.

- **Flow Control**
  - Prevents sender from overwhelming the receiver
  - If buffer overflows:
    - Drop frames or
    - Send feedback to pause/slow down sender
  - *Used also at transport layer*

Services of Data Link Layer (2)

- **Error Control**
  - Handles errors from physical transmission
  - Steps:
    - Detect errors
    - Correct at receiver or
    - Request retransmission
  - Critical for reliable communication
- **Congestion Control**
  - Congestion in link results in frame loss
  - Rarely implemented in data-link layer
    - Handled mostly in network & transport layers

Two Sublayers

- **Data Link Control (DLC) Sublayer**
  - Handles functions **common to all link types**
    (e.g., framing, flow control, error control)
  - Used in both **point-to-point** and **broadcast** links
- **Media Access Control (MAC) Sublayer**
  - Deals with issues **specific to broadcast links**
    (e.g., controlling access to a shared medium)
  - Crucial in **LANs** and wireless networks

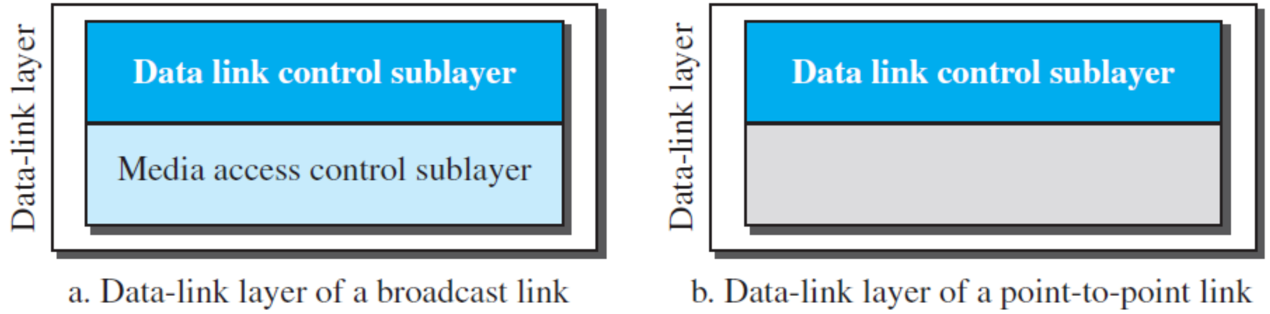


Figure 9. 4: Dividing the data-link layer into two sublayers

Link Layer Addressing

- IP addresses identify source and destination hosts in the Internet
- Internet is connectionless. Datagrams may take different paths.
- IP addresses must not change during transmission

- Thus Link Layer addressing is used deliver frames over each individual link.

Link Layer Addressing

- Link Layer Address is also known as MAC address, Physical address.
- Unique **per node on a link.**

Every time a datagram is passed to a new link, **new link-layer addresses** are assigned in the frame, even though the **IP addresses remain unchanged**.

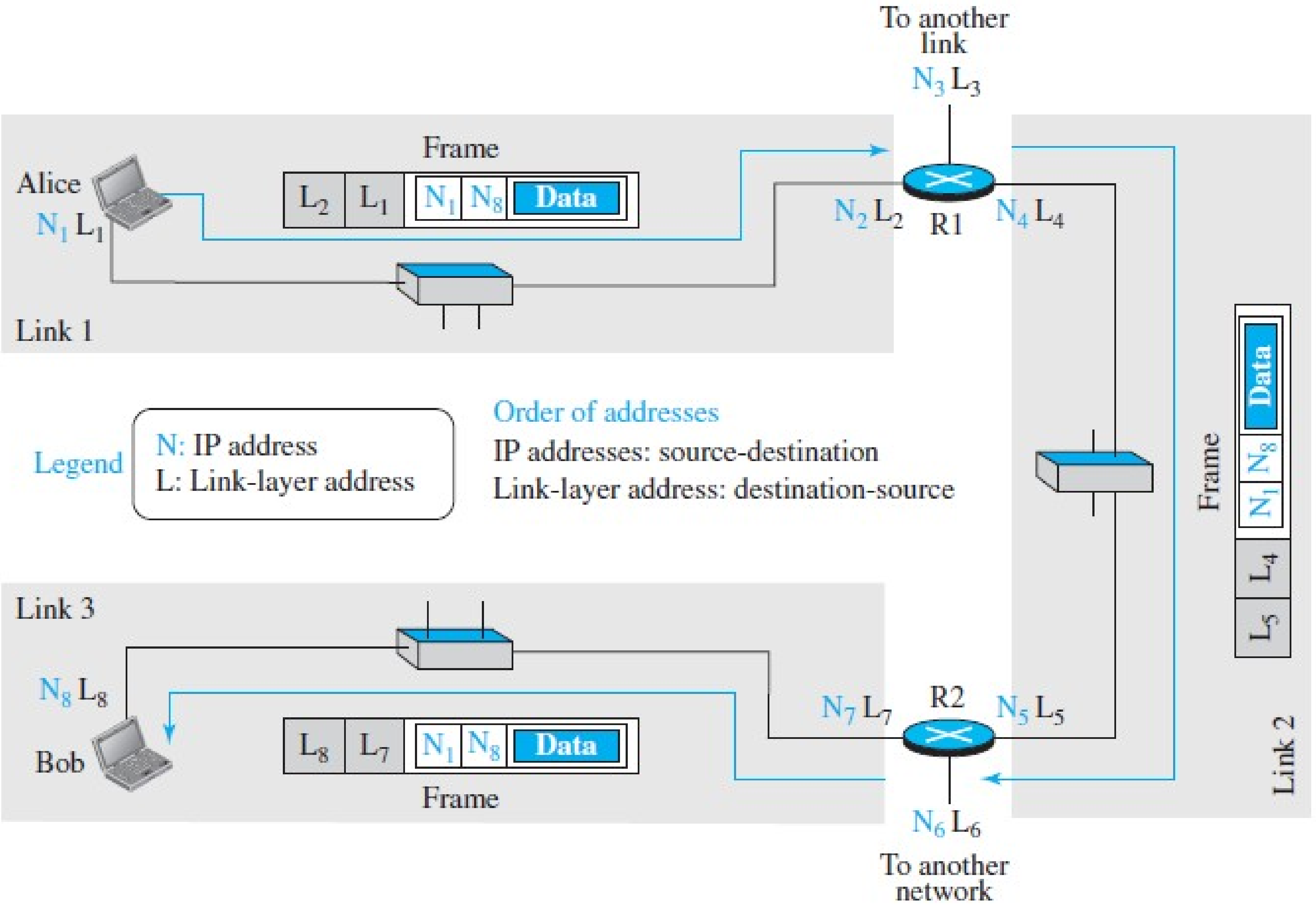


Figure 9. 5: IP addresses and link-layer addresses in a small internet

Link Layer Addressing

- Q1:Why do routers need IP addresses if they don't appear in datagrams?
  - Routers do appear as senders/receivers in routing protocols (e.g., RIP, OSPF)
- Q2: Why does each router interface need a separate IP address?
  - Each interface is connected to a different link
- Q3: How are source & destination IP addresses set?
  - Source host uses Own IP as source, Destination IP found via DNS (Domain Name System)
- Q4: How are link-layer addresses determined per link?
  - Each node knows its own MAC address
  - Destination MAC address is obtained via ARP (Address Resolution Protocol)
- Q5: What is the size of link-layer addresses?
  - Depends on the data-link protocol used (e.g. Ethernet: 48 bits and Other links may use different sizes)

Types of Link-Layer Addresses

- **Unicast Address**
  - Assigned to **a single host or router interface**
  - Enables **one-to-one** communication
  - Frame is delivered to **one specific receiver**

00-1d-a1-02-b3-c4
Unicast Address

- **Multicast Address**
  - Represents a **group of devices** on the same link
  - Enables **one-to-many** communication
  - Frame is delivered to **all members of the group**
  - Scope is **local to the link**

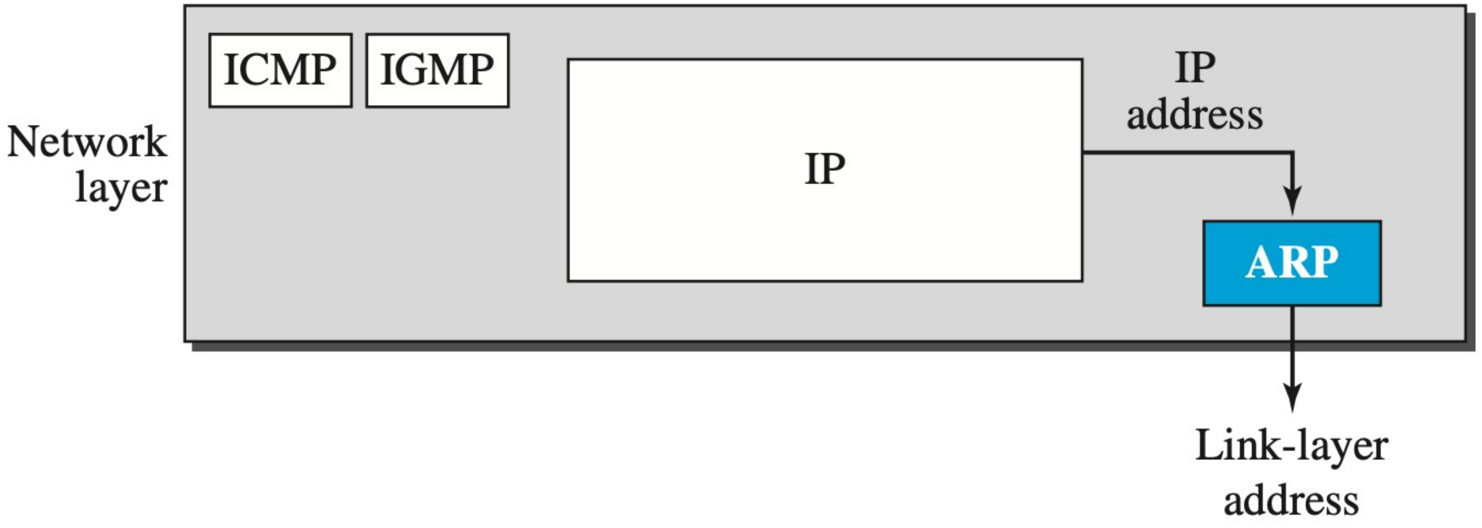
01-00-5e-00-00-16
Multicast Address

- **Broadcast Address**
  - Special address used to reach **all devices** on the link
  - Enables **one-to-all** communication
  - Frame is received by **every node** in the network segment

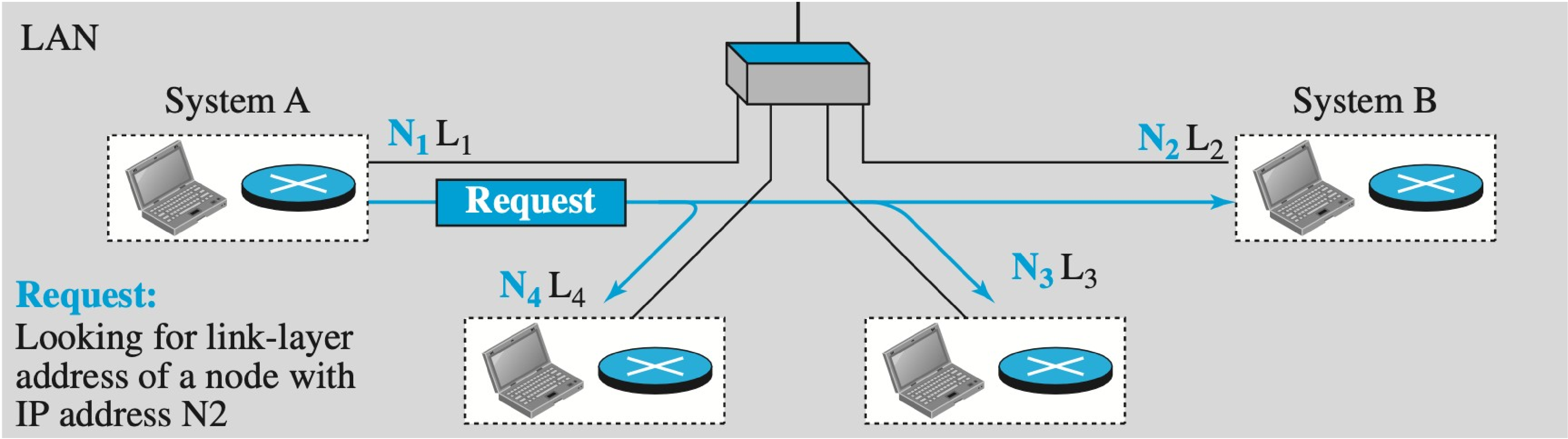
ff-ff-ff-ff-ff-ff
Broadcast Address

Address Resolution Protocol (ARP)

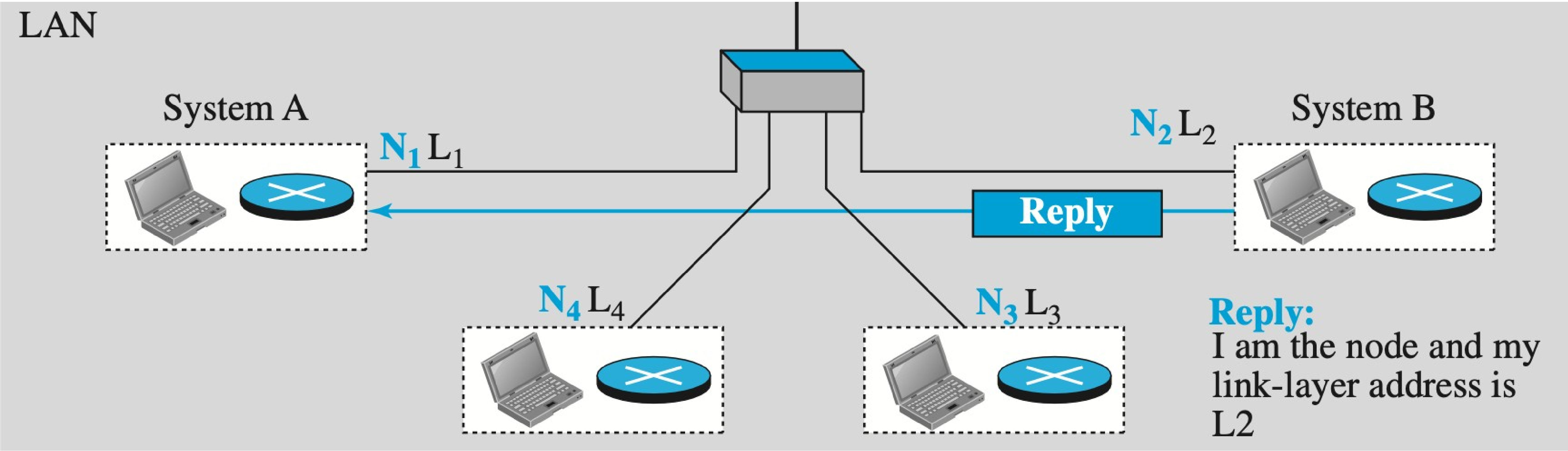
- ARP table has a mapping between IP address vs. Link Layer Address.
- This table is populated by ARP protocol.
- ARP is an auxiliary protocol. Its Layer 3 protocol.



ARP Request


LAN
System A
N_1 L_1
System B
N_2 L_2
Request
N_4 L_4
N_3 L_3
Request:
Looking for link-layer address of a node with IP address N2

ARP Reply


LAN
System A
N_1 L_1
N_2 L_2
System B
Reply
N_4 L_4
N_3 L_3
Reply:
I am the node and my link-layer address is L2

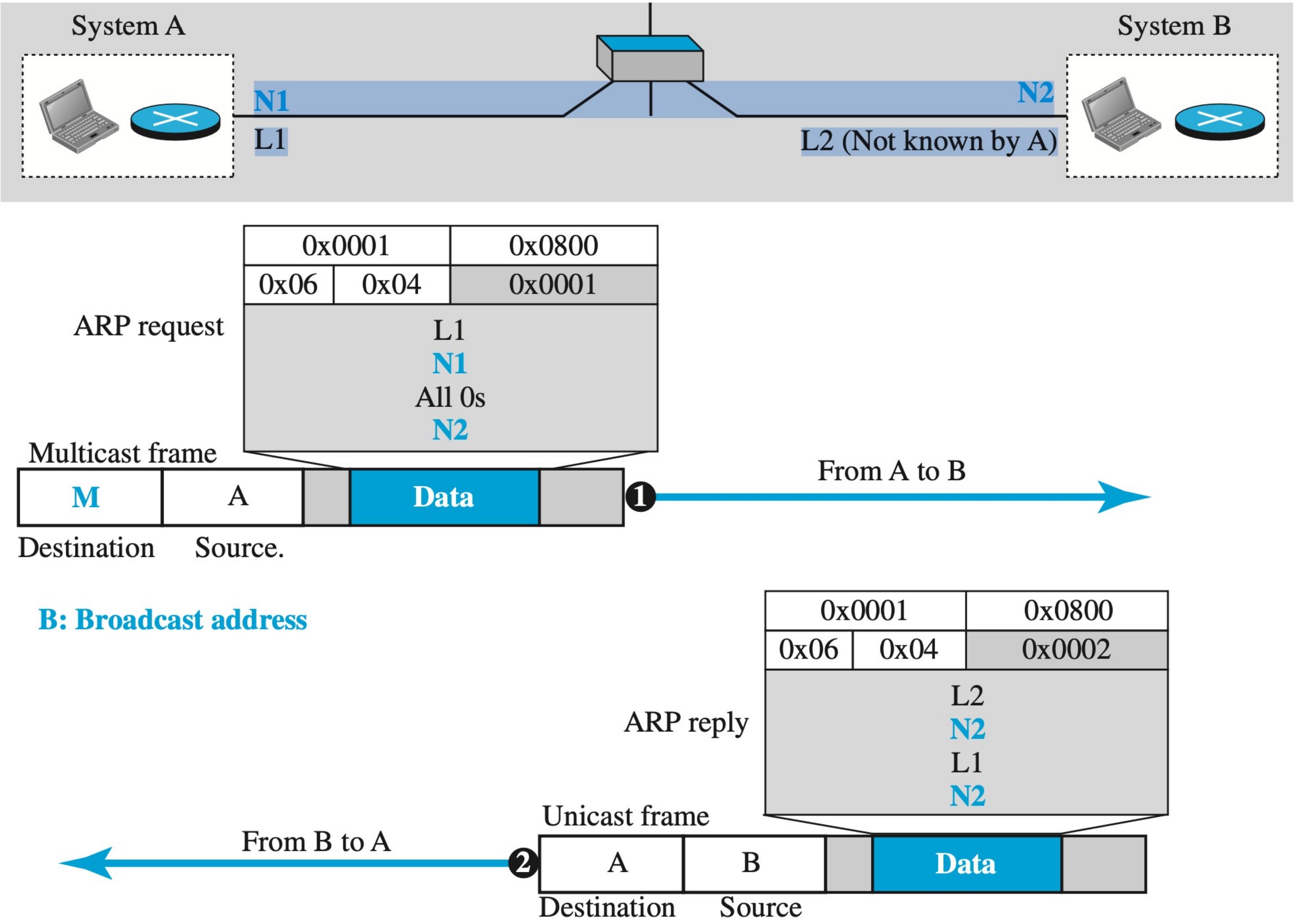
ARP Packet Format

0	8	16 31
Hardware Type		Protocol Type
Hardware length	Protocol length	Operation **Request:1, Reply:2**
Source hardware address		
Source protocol address		
Destination hardware address (Empty in request)		
Destination protocol address		

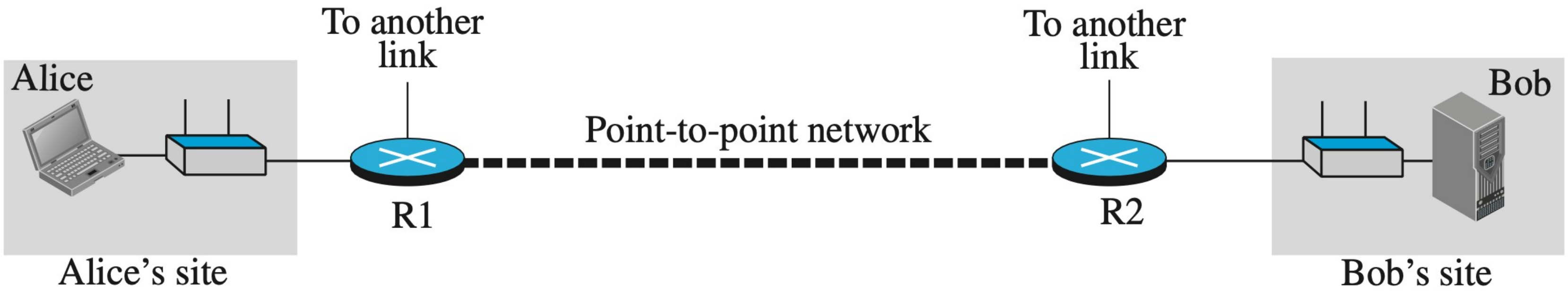
Hardware: LAN or WAN protocol

Protocol: Network-layer protocol

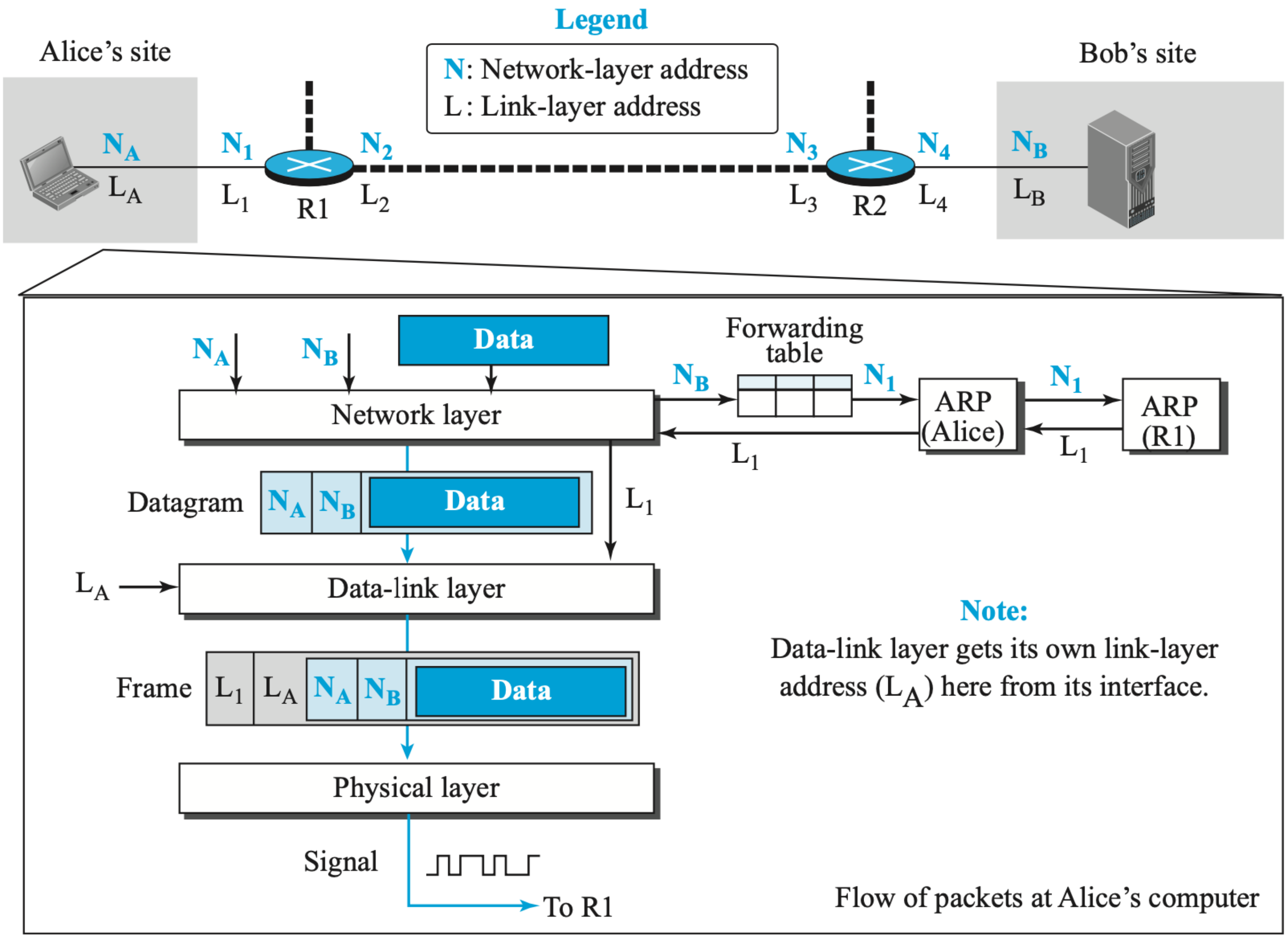
ARP Packet Example



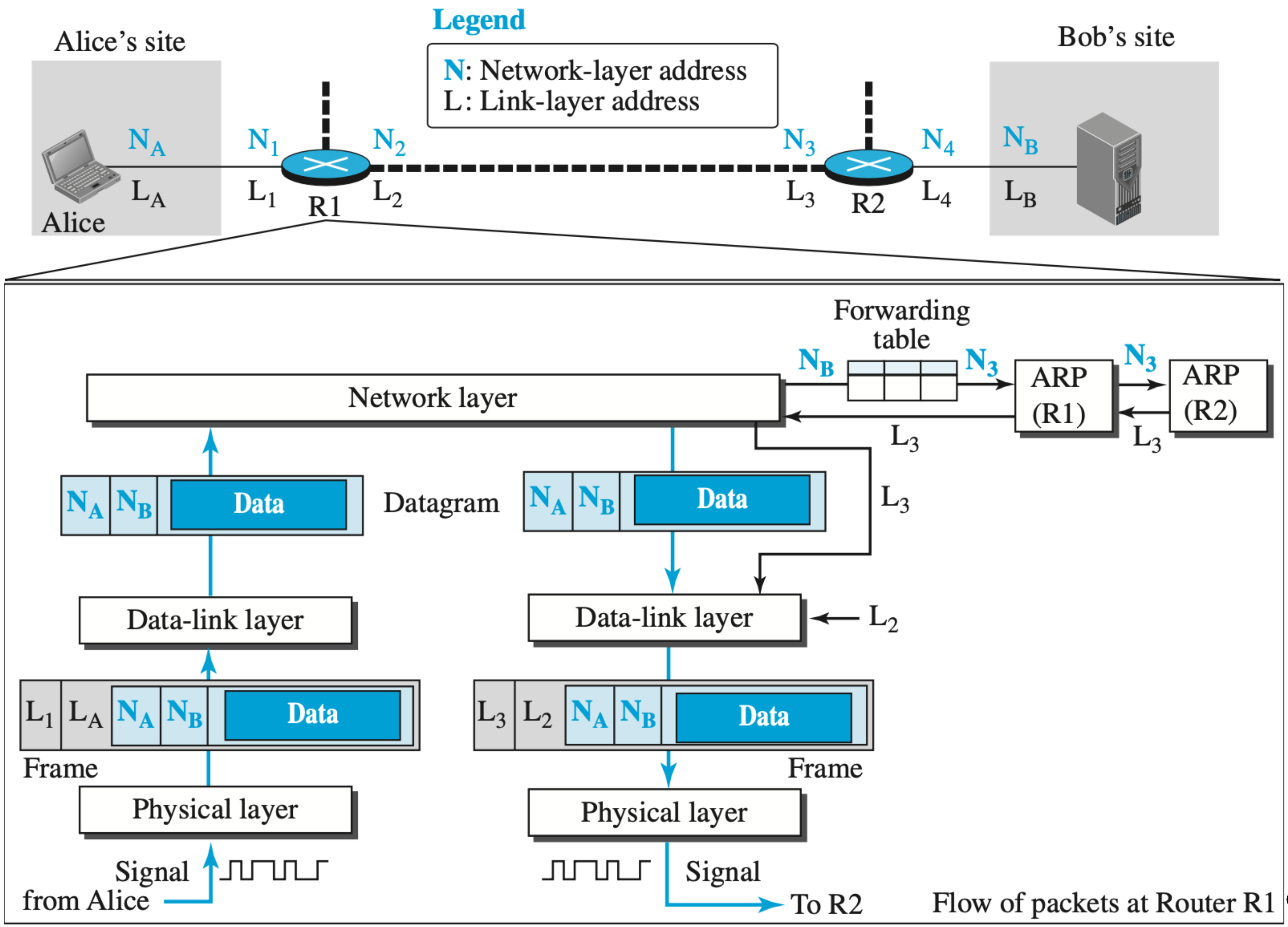
An Example of Communication



Activities at Alice's Site

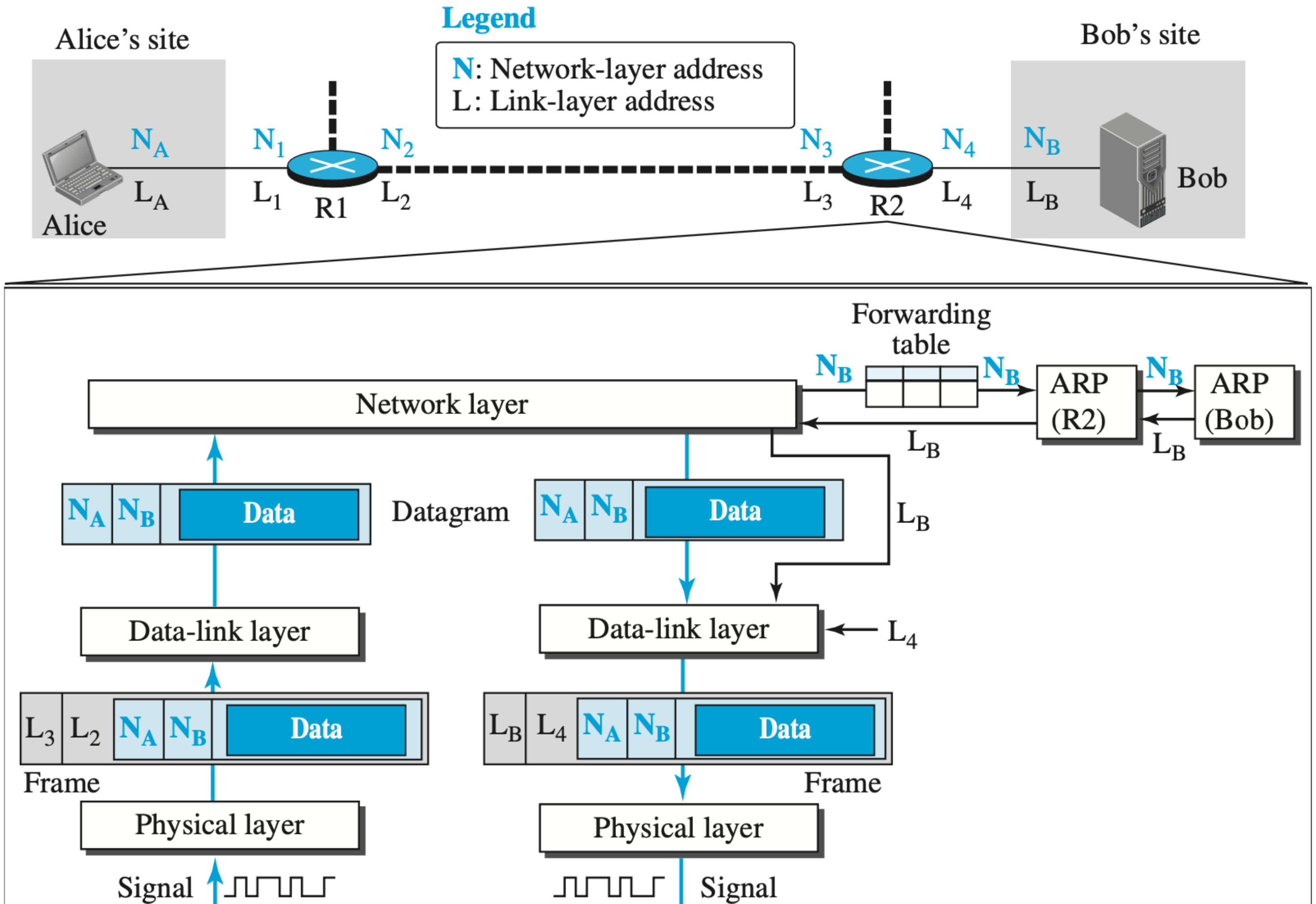


Activities at Router R1



Flow of packets at Router R1

Activities at Router R2



Flow of packets at Router R2

Activities at Bob’s Site